FINAL INSIGHTS DATA MODELLING &TRANSFORMATION USING POWER BI

Project Title: **INVENTORY SALES ANALYSIS**

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Dataset: Ecommerce retail dataset. In this dataset we have total six table:

1) List of Orders

2) Order Details

3) Sales Target

4) Orders Data

5) Order Details Duplicate

6) Sales Target Duplicate

Key fields:

Order Id, Order date, Customer name, State, City, Location, Category, Sub-Category, Amount, Profit, Quantity, Profit Margin, Profit status, Order Count, Average Profit, Total Amount, Month of Order Date, Target, Month, Total Target.

Data Preparation: Data import, Power Query transformations

● Data Cleaning: Formatting, null handling, deduplication

Handling Missing Data & Duplicate Data

CLEAN FUNCTION: Removes line breaks, charterers and other hidden control.

TRIM FUNCTION: This removes non printable characters and extra spaces at the same time.

PROPER FUNCTION: The text can replace capitalized each word.

NULL FUNCTION: you can replace null values using find and replace using replace values options.

For sales data, you can use Power Query to remove blanks, standardize customer names, group by order data using count,average,sum fuctions, and calculate total Amount before loading the result into Excel. That makes reporting much easier and more reliable.

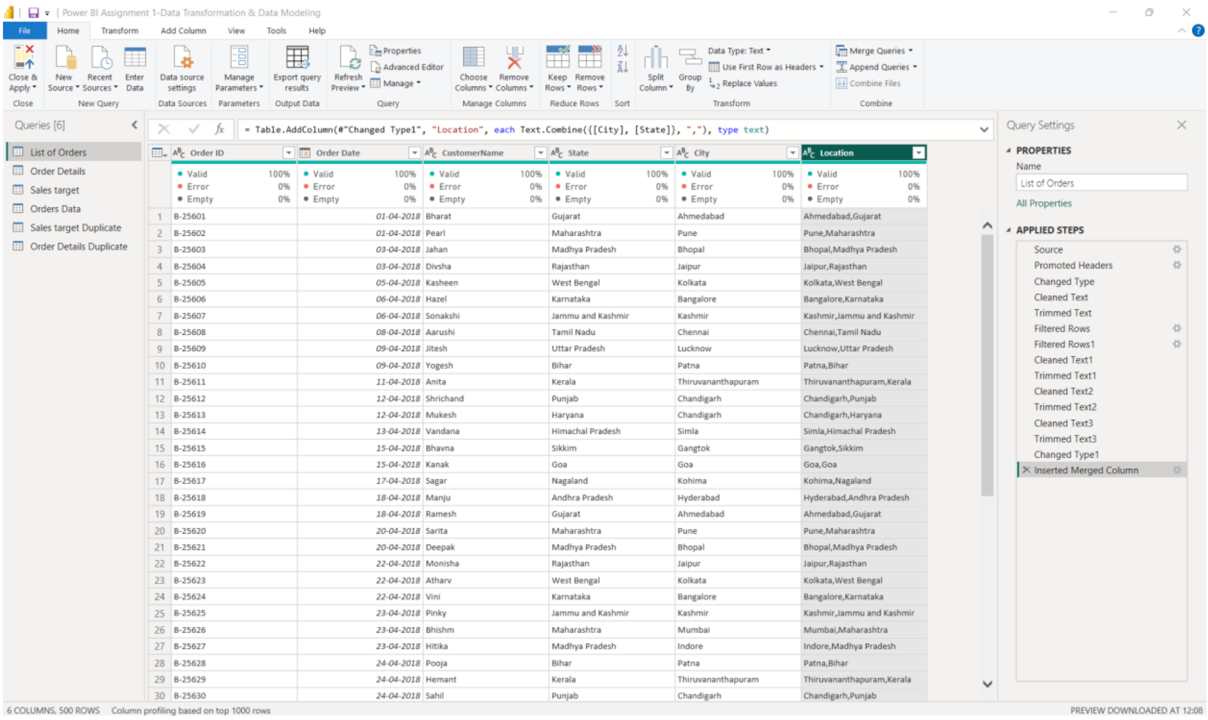
● Data Integration: Table merging and relationships

Column Creation:

1) Merge the City and State columns to create a new column named Location in the format: ***City, State***

Screenshot of Location Table:

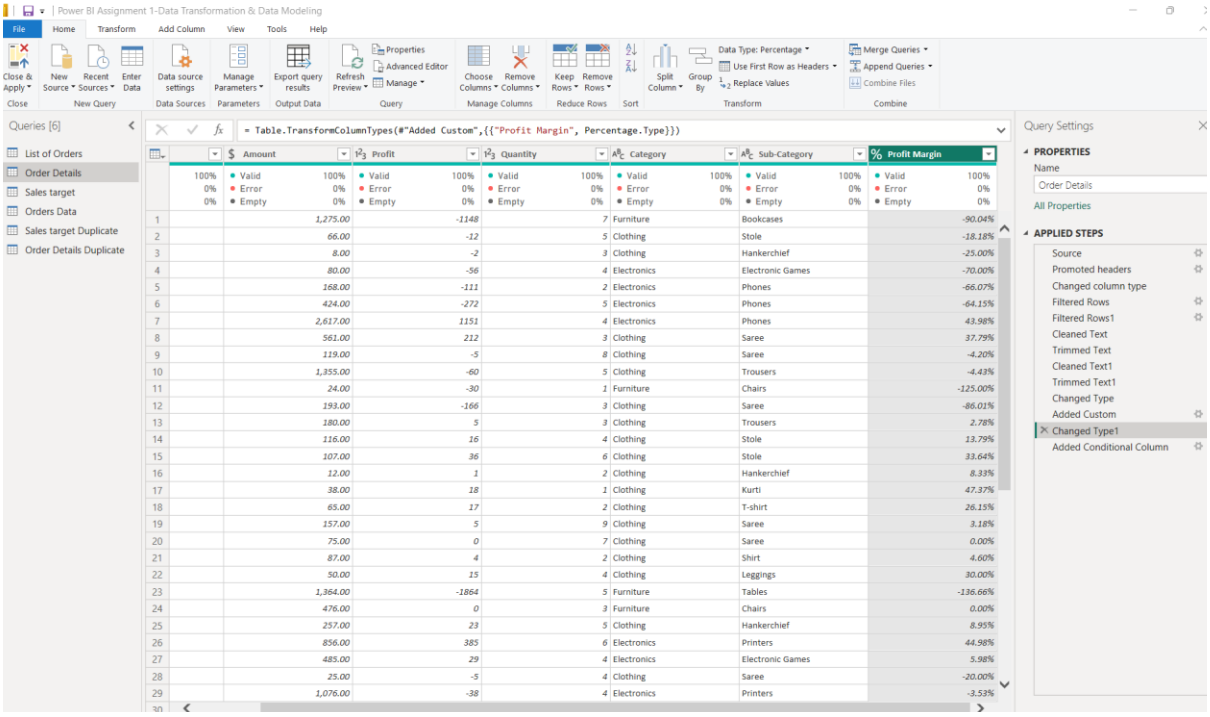
Answer:



2)Create a custom column named **Profit Margin** calculated as: ***Profit / Amount*** (expressed as a percentage)

Screenshot of Profit Margin Table:

Answer:



	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
1	1,275.00	-1148	7	Furniture	Bookcases	-90.04%
2	66.00	-12	5	Clothing	Stole	-18.18%
3	8.00	-2	3	Clothing	Hankerchief	-25.00%
4	80.00	-56	4	Electronics	Electronic Games	-70.00%
5	168.00	-111	2	Electronics	Phones	-66.07%
6	424.00	-272	5	Electronics	Phones	-64.15%
7	2,617.00	1151	4	Electronics	Phones	43.98%
8	561.00	212	3	Clothing	Saree	37.79%
9	119.00	-5	8	Clothing	Saree	-4.20%
10	1,355.00	-60	5	Clothing	Trousers	-4.43%
11	24.00	-30	1	Furniture	Chairs	-125.00%
12	193.00	-166	3	Clothing	Saree	-86.01%
13	180.00	5	3	Clothing	Trousers	2.78%
14	116.00	16	4	Clothing	Stole	13.79%
15	107.00	36	6	Clothing	Stole	33.64%
16	12.00	1	2	Clothing	Hankerchief	8.33%
17	38.00	18	1	Clothing	Kurti	47.37%
18	65.00	17	2	Clothing	T-shirt	26.15%
19	157.00	5	9	Clothing	Saree	3.18%
20	75.00	0	7	Clothing	Saree	0.00%
21	87.00	4	2	Clothing	Shirt	4.60%
22	50.00	15	4	Clothing	Leggings	30.00%
23	1,364.00	-1864	5	Furniture	Tables	-136.66%
24	476.00	0	3	Furniture	Chairs	0.00%
25	257.00	23	5	Clothing	Hankerchief	8.95%
26	856.00	385	6	Electronics	Printers	44.98%
27	485.00	29	4	Electronics	Electronic Games	5.98%
28	25.00	-5	4	Clothing	Saree	-20.00%
29	1,076.00	-38	4	Electronics	Printers	-3.53%

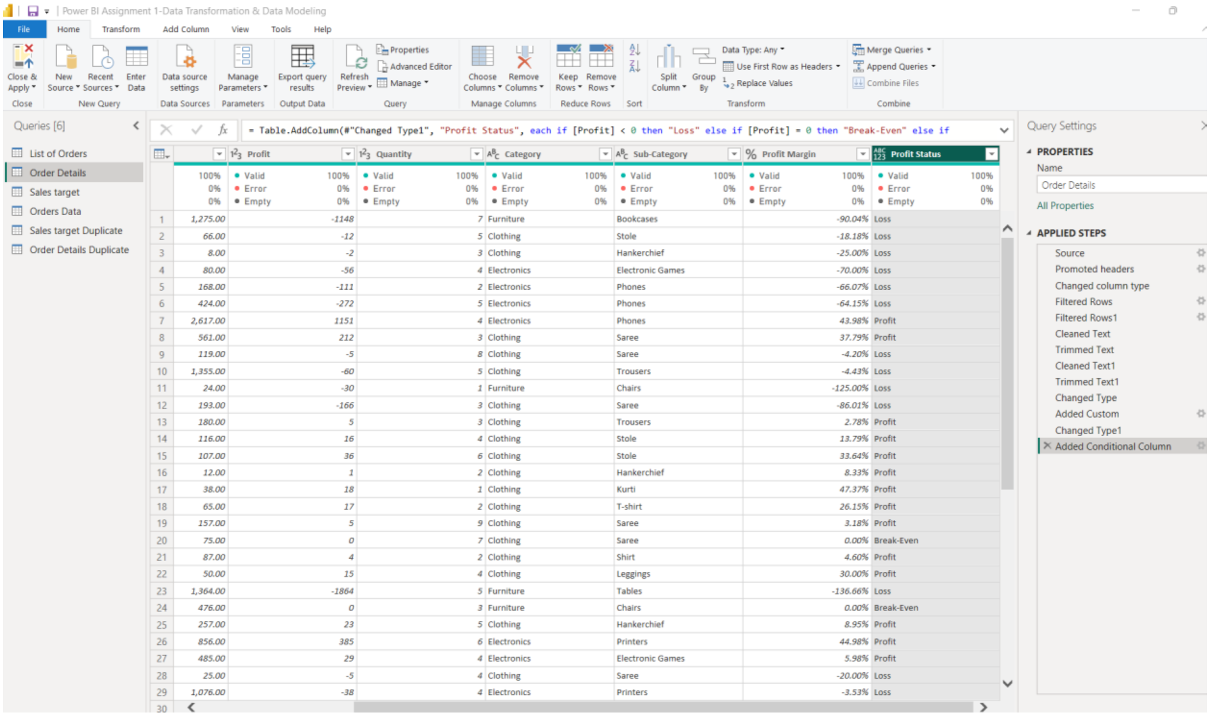
Conditional Column:

1)Add a conditional column named **Profit Status** based on: ■ Profit < 0 → *Loss* ■ Profit = 0 → *Break-Even*

■ Profit > 0 → *Profit*

Screenshot of Profit Status Table:

Answer:

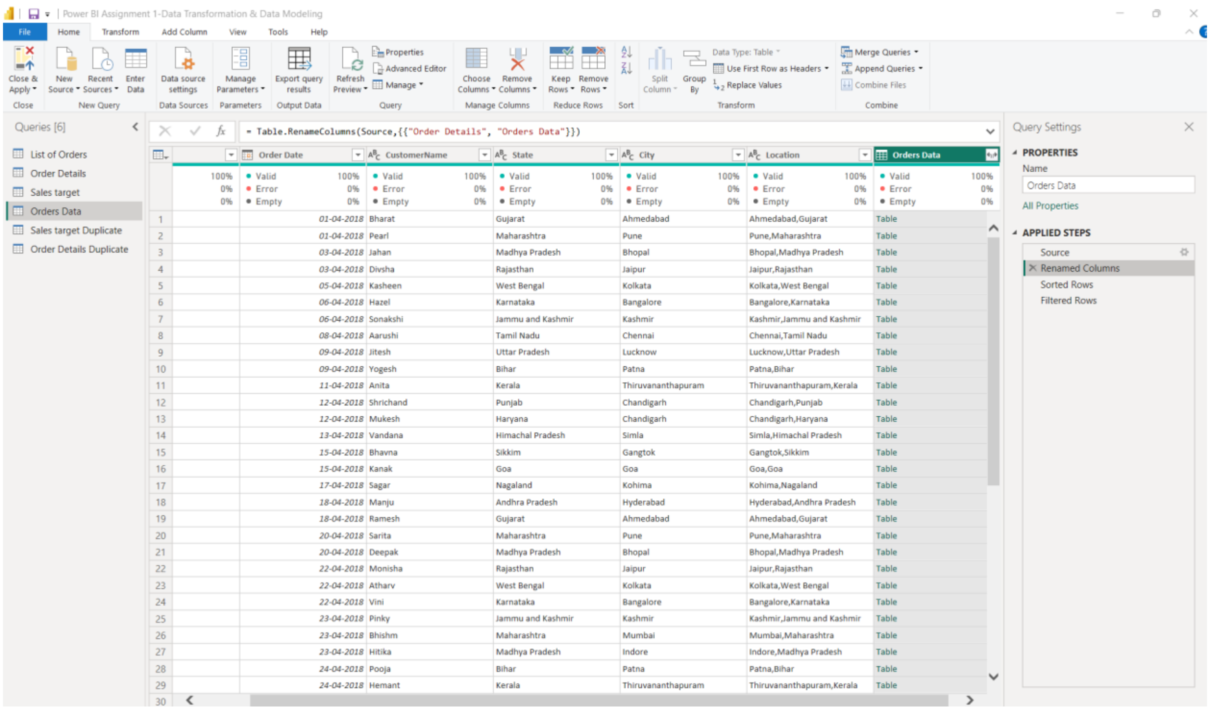


Merging Data (Joins)

- Merge the *List of Orders* and *Order Details* tables using **Order ID**.
- Name the merged output table as **Orders Data**.

Screenshot of Orders Data Table:

Answer:

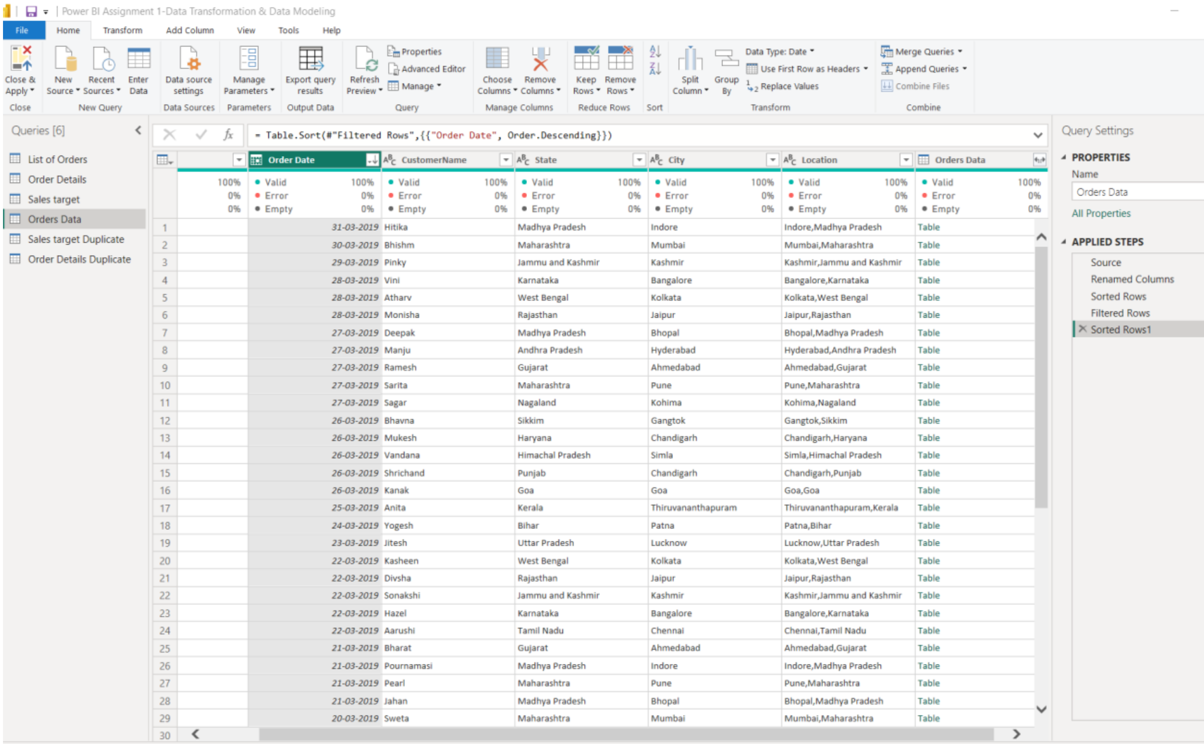


8. Sorting and Filtering Data

1. Sort records in the **Orders Data** table by **Order Date** in **descending order**.

Screenshot of Table:

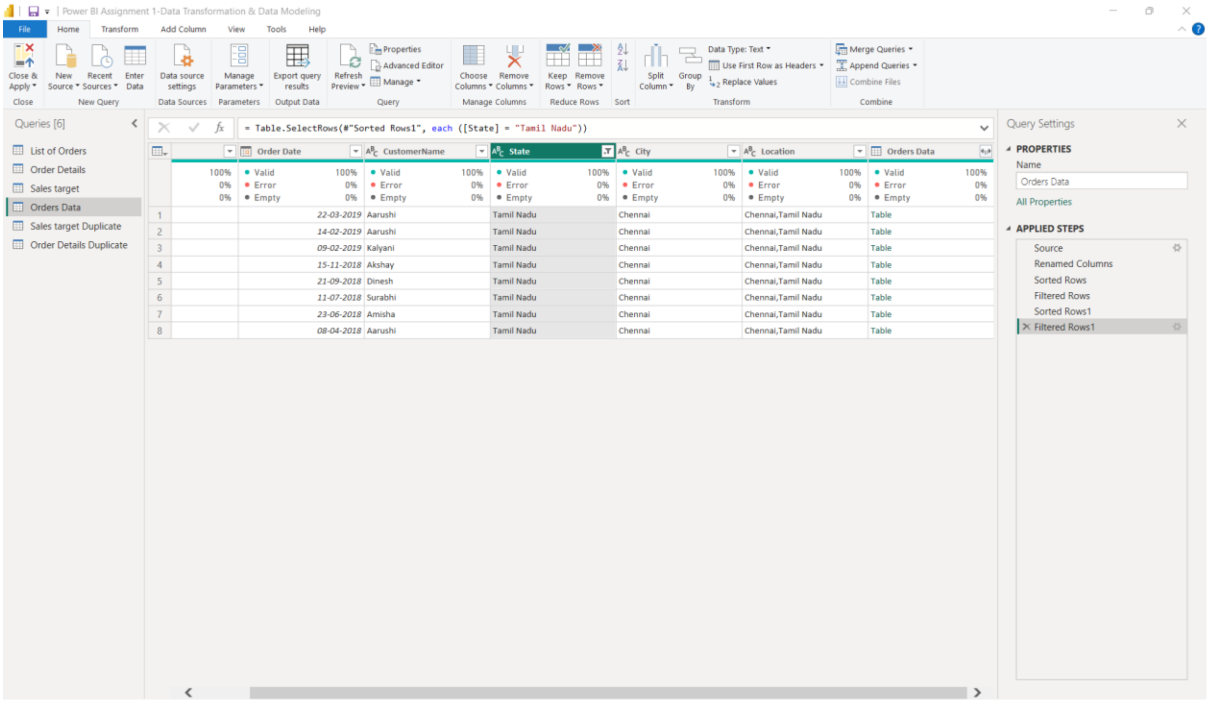
Answer:



2. Apply filters to analyse orders from a specific state (e.g., **Tamil Nadu**).

Screenshot of Table:

Answer:



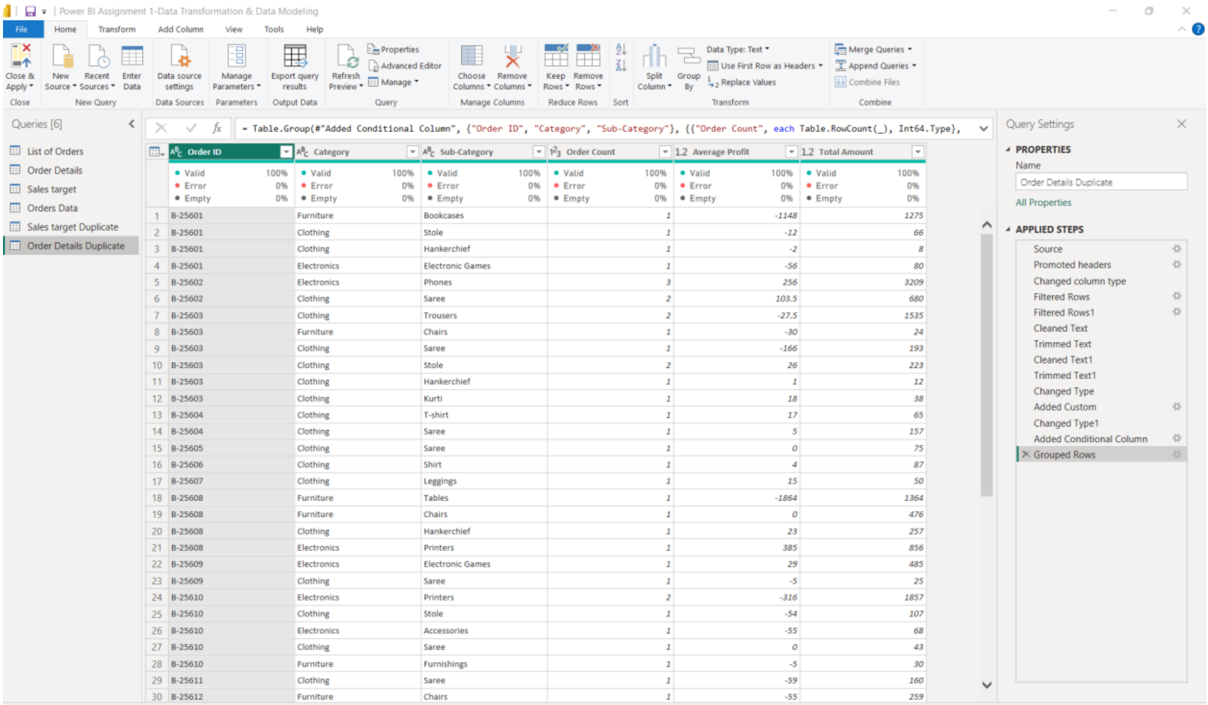
● Data Aggregation: Grouping and summarization :

Data aggregation is the process of combining detailed records into summaries based on

- Count of Order ID
- Average Profit by Category
- Total Amount by Sub-Category

<u>Screenshot of Table:</u>

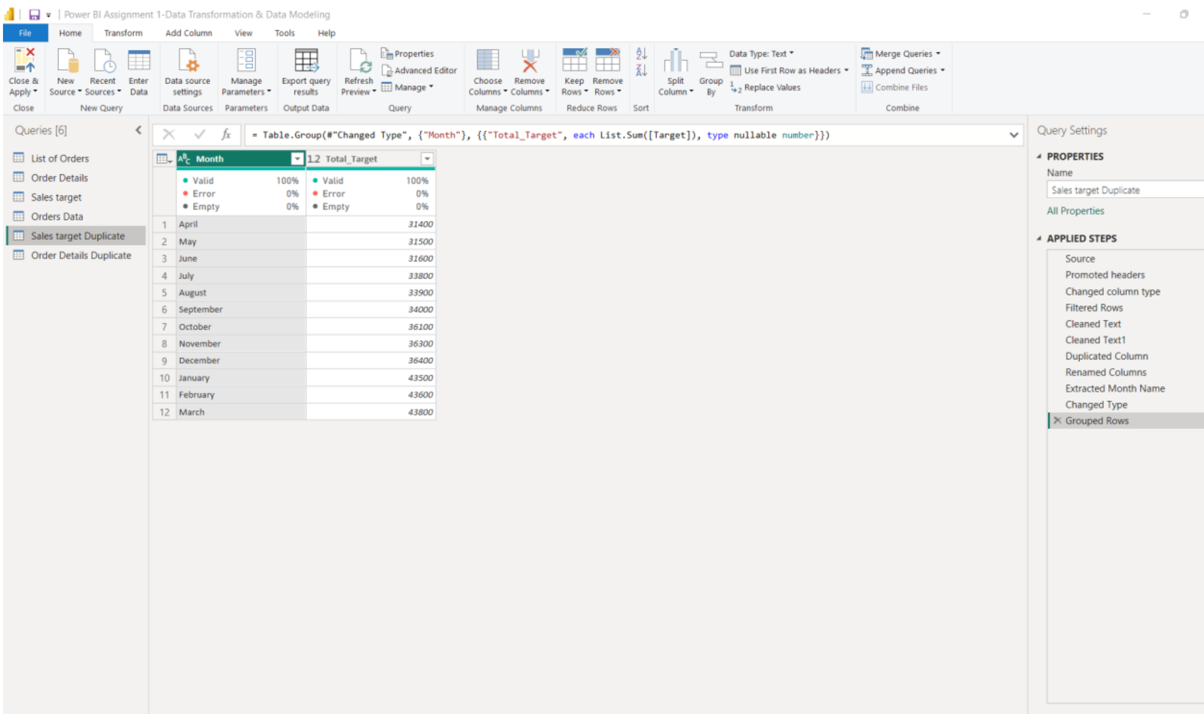
<u>Answer:</u>



➢ Duplicate the *Sales Target* table and aggregate the total Target by **Month of order date.**

<u>Screenshot of Table:</u>

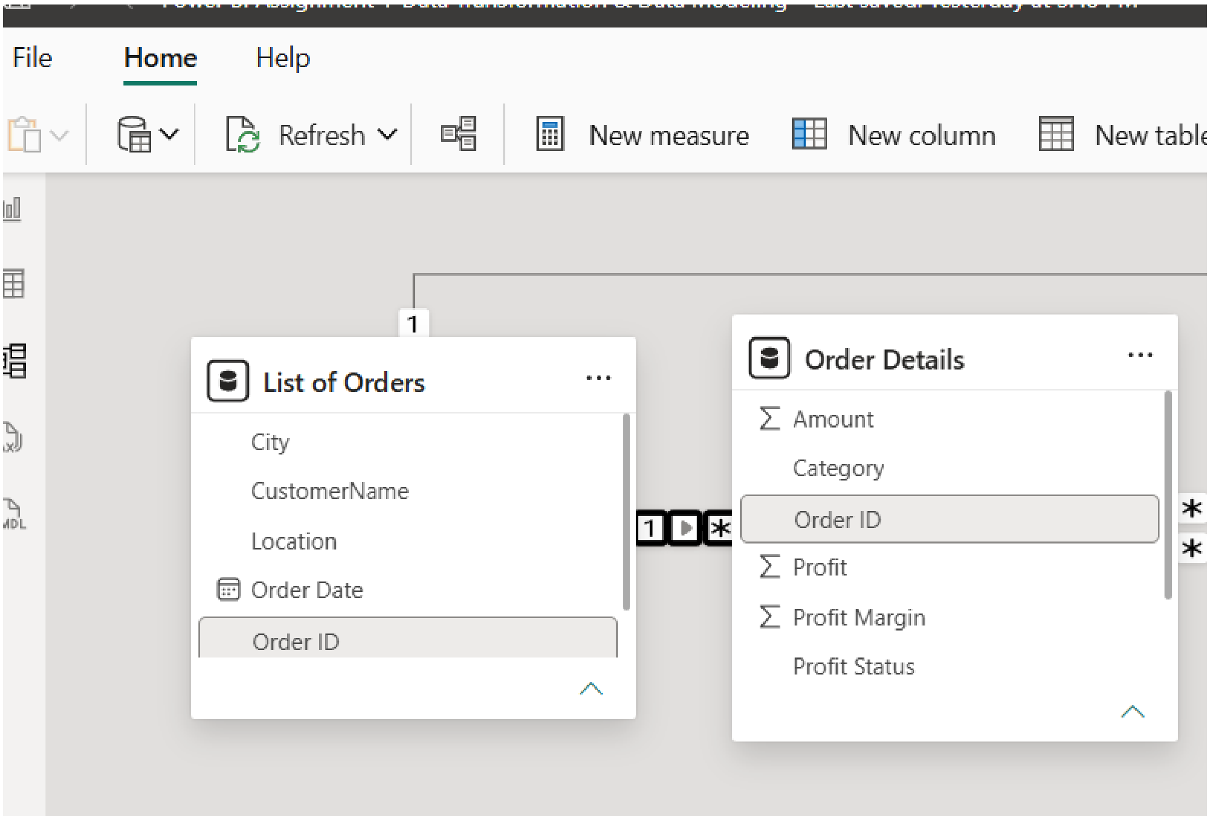
<u>Answer:</u>



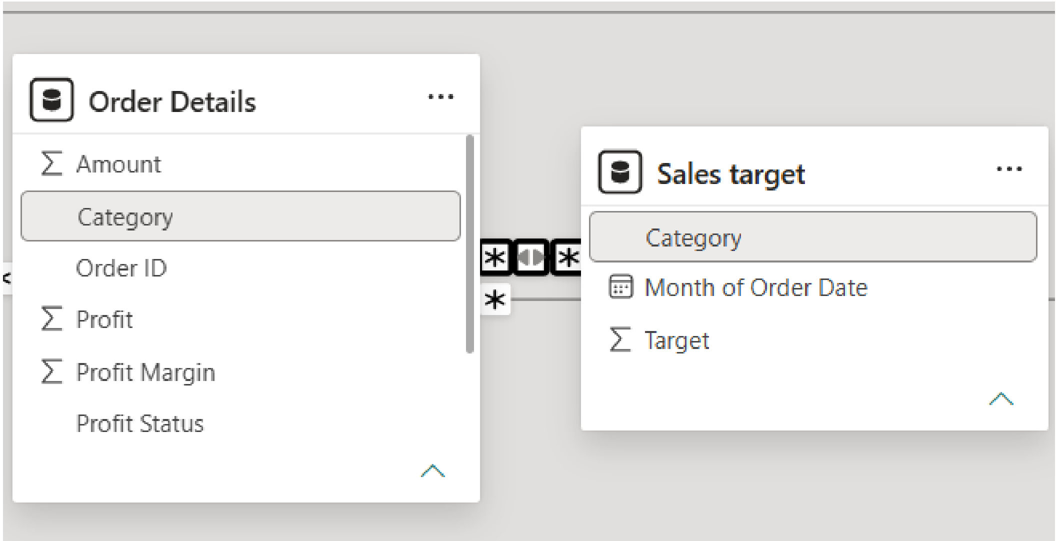
● Data Modeling: Schema design and relationship management.

1)Establish a relationship between:

- *List of Orders* and *Order Details* using **Order ID**



1) Establish a relationship between:
   - *Order Details* and *Sales Target* using **Category**



➢ Use **Manage Relationships** to ensure relationships are active and correctly defined.